MedFlow AI Cybersecurity Ecosystem

Architecture Summary

Project Purpose

An AI-powered red team and threat intelligence platform for healthcare cybersecurity simulations, combining LLM agents with RAG (Retrieval-Augmented Generation) to simulate attacks, detect threats, and provide SOC-ready defensive guidance.

1. Core Architecture Layers

CLI Interface (app.py) → Commands: ask, search, compare, investigate

Agent Layer: LLM-based RAG agents for threat intelligence
Graph Layer: Stateful investigation pipeline for alert processing

Data Layer: Vector databases (Chroma) + External integrations (Wazuh, MISP, MITRE ATT&CK;)

2. Four Specialized AI Agents (RAG-Based LLM)

Agent	Role	Vector DB	Use Case
Red Team	Attack planning & ATT&CK mapping	playbook_db	Kill-chain generation for healthcare
CTI	Cyber threat intelligence lookup	attack_db	Technique/actor/campaign information
Attribution	Threat actor attribution	actor_db	Links IOCs/behaviors to known actors
Detection	SIEM rules & detection engineering	detection_db	Detection logic for healthcare threats

Key Feature: All agents use grounding prompts that prioritize retrieved facts over general knowledge, label verified vs. unverified claims, and output source summaries.

3. Threat Intelligence Graph (Stateful Investigation Pipeline)

Technology: LangGraph state machine

Pipeline Flow:
Collection → IOC Enrichment → [Confidence Check] → MITRE Lookup → Correlation → Prediction → Reporting → END

Key Nodes:

• **Collection:** Pulls alerts from Wazuh (live) or simulated scenarios (JSON files)

• **IOC Enrichment:** Enriches indicators via MISP API (optional) or local context

• **MITRE Lookup:** Maps IOCs to MITRE ATT&CK; techniques

• **Correlation:** Groups related techniques into campaigns

• **Prediction:** Forecasts next attacker steps using LLM

• **Reporting:** Generates formatted investigation summary

4. Vector Database Architecture (Chroma)

Database	Built From	Purpose	Query Method
attack_db	Enterprise ATT&CK JSON	Technique/tactic/procedure lookup	Semantic search on descriptions
redteam_db	ATT&CK + CAPEC	Offensive technique context	Kill-chain/MITRE technique queries
actor_db	Actor profiles & campaigns	Threat actor attribution	Behavior/IOC-to-actor mapping
detection_db	ATT&CK detection guidance	SIEM/detection rules	Data source/technique-to-detection

Embedding Model: Sentence transformers (configurable, cached locally)

5. Data Sources

• MITRE CTI: Enterprise ATT&CK;, CAPEC, mobile/ICS/pre-attack frameworks

• Wazuh Integration: Live alert streaming from SIEM (optional)

• MISP Integration: IOC enrichment via local OSINT feeds (optional Docker deployment)

• Simulated Scenarios: JSON alert files (e.g., Ryuk ransomware labs)

6. Healthcare-Focused Scenarios

• MFA fatigue attacks

• API abuse

• Phishing campaigns

• Ransomware staging (Ryuk)

• Smart-contract permission abuse

7. LLM Configuration

Provider: Groq (fast inference)
Main Model: Configurable (default from .env)
Secondary Model: For comparison/validation
Model Caching: Hugging Face models cached locally for offline use

Key Features:
• Model comparison: Run same query on main vs. secondary
• Grounding: All answers anchored to retrieved data with source tracking

8. Deployment Options

• Standalone: Python CLI (no external deps required)

• With MISP: Optional Docker deployment for local threat feed enrichment

• With Wazuh: Optional Docker deployment for live SIEM integration

9. Key Workflows

Query Workflow:

User Question → RAG Agent → Vector DB Retrieval → LLM Generation → Grounded Answer

Investigation Workflow:

Scenario (ryuk/live) → Alert Collection → IOC Enrichment → MITRE Mapping → Campaign Correlation → Prediction → HTML Report

Setup Workflow:

Download MITRE CTI → Build Vector DBs → Configure Groq API → Validate Setup → Ready for Queries/Investigations

10. Key Technologies

• **Framework:** LangGraph (agentic state machines)

• **Vector Store:** Chroma (local, embeddable)

• **Embeddings:** Sentence Transformers (offline)

• **LLM:** Groq API (fast, low-latency)

• **Data Format:** JSON (MITRE), Wazuh alerts (JSON)

• **Optional:** MISP Docker, Wazuh Docker, stunnel (SSL tunneling)

Summary

This architecture enables **grounded threat intelligence** where every AI answer is traceable to verified security data (MITRE ATT&CK;, actor profiles, detection techniques), making it suitable for healthcare SOC training and red team simulations.

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